Performance Evaluation

COMP7404

Computational Intelligence and Machine Learning

Dong Xu

Assessment

- 2 Assignments (15 %)
- Group project (35 %)
  - Read and implement machine learning research paper
  - Presentation to class (8 minutes presentation)
  - Max 36 Groups
  - 4 students per group for most groups (~140 students enrolled in this class)
- Final examination (50 %): written, closed book, 2 hours

Assessment - Project - Group Selection

- Select your group on Moodle
  - Start: 26 Jan
  - Deadline: 5 Feb 23:55
  - We will randomly assign a group for you if you don't make a selection by the deadline
- Member swapping
  - Allowed if approved before start of project
    - All group members of both groups must agree

Assessment - Project - Paper Selection

- Read a machine learning paper
- Implement / test the paper
    - May use CS GPU Farm
    - Employ own test/training data if possible
    - Make your code available to others via github
- List of project topics will be made available on 27 Sep
- Topic selection
    - Start on 6 Feb
    - Deadline: 11 Feb 23:59

A few possible papers from my research group

- F. Nie*, Z. Zeng*, I. W. Tsang, **D. Xu** and C. Zhang, "Spectral Embedded Clustering: A Framework for In-Sample and Out-of-Sample Spectral Clustering," *IEEE Transactions on Neural Networks* (T-NN), 22(11), pp. 1796-1808, November 2011.
- Y. Yang*, **D. Xu**, F. Nie*, S. Yan and Y. Zhuang, "Image Clustering using Local Discriminant Models and Global Integration," *IEEE Transactions on Image Processing* (T-IP), 19(10), pp. 2761-2773, October 2010.
- **D. Xu**, S. Yan, L. Zhang, S. Lin, H. Zhang and T. S. Huang, "Reconstruction and Recognition of Tensor-based Objects with Concurrent Subspaces Analysis," *IEEE Transactions on Circuits and Systems for Video Technology* (T-CSVT), 18(1), pp. 36-47, January 2008.
- S. Yan, **D. Xu**, Q. Yang, L. Zhang, X. Tang and H. Zhang, "Multilinear Discriminant Analysis for Face Recognition," *IEEE Transactions on Image Processing* (T-IP), 16(1), pp. 212-220, January 2007.
- **D. Xu**, S. Yan, D. Tao, L. Zhang, X. Li and H. Zhang, "Human Gait Recognition with Matrix Representation," *IEEE Transactions on Circuits and Systems for Video Technology* (T-CSVT), 16(7), pp. 896-903, July 2006.
- L. Duan*, I. W. Tsang and **D. Xu**, "Domain Transfer Multiple Kernel Learning," *IEEE Transactions on Pattern Analysis and Machine Intelligence* (T-PAMI), 34(3), pp. 465-479, March 2012.
- L. Duan*, **D. Xu** and I. W. Tsang, "Domain Adaptation from Multiple Sources: A Domain-Dependent Regularization Approach," *IEEE Transactions on Neural Networks and Learning Systems* (T-NNLS), 23(3), pp. 504-518, March 2012.
- W. Li*, L. Duan*, **D. Xu** and I. W. Tsang, "Learning with Augmented Features for Supervised and Semi-supervised Heterogeneous Domain Adaptation," *IEEE Transactions on Pattern Analysis and Machine Intelligence* (T-PAMI), 36(6), pp. 1134-1148, June 2014.
- W. Li*, Z. Xu*, **D. Xu**, D. Dai and L. Van Gool, "Domain Generalization and Adaptation using Low Rank Exemplar SVMs," *IEEE Transactions on Pattern Analysis and Machine Intelligence* (T-PAMI), 40(5), pp. 1114-1127, May 2018.

Outline of Lecture

- Performance Evaluation

Precision, Recall, and F-measure (1)

- Suppose the cutoff threshold is chosen to be 0.8. In other words, any instance with posterior probability greater than 0.8 is classified as positive.
- Compute the precision, recall, and F-measure for the model at this threshold value.

Instance	P(+|A)	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.852	-
5	0.851	-
6	0.850	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

Precision, Recall, and F-measure (2)

	PREDICTED CLASS		
ACTUAL CLASS		Class =Yes	Class= No
	Class =Yes	(TP) 3	(FN) 2
	Class =No	(FP) 3	(TN) 2

Instance	P(+|A)	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.852	-
5	0.851	-
6	0.850	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

Precision, Recall, and F-measure (3)

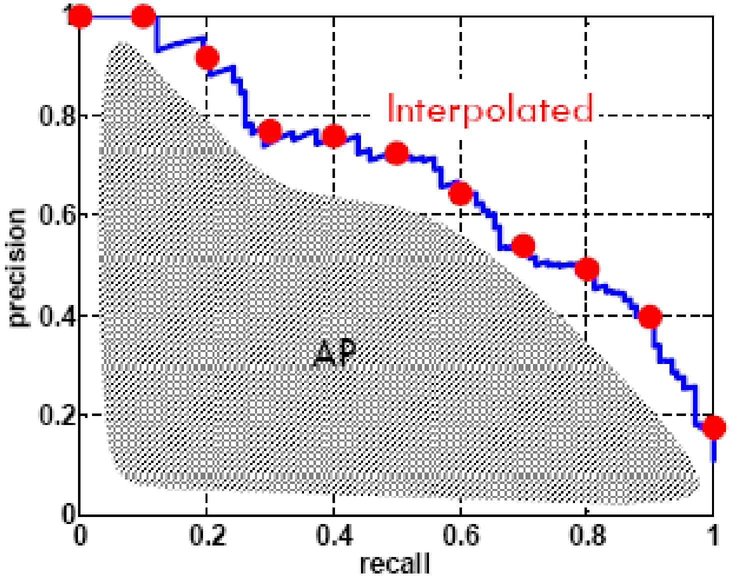
	PREDICTED CLASS		
		Class =Yes	Class= No
ACTUAL CLASS	Class =Yes	(TP) 3	(FN) 2
	Class =No	(FP) 3	(TN) 2

p = TP/(TP+FP)= 3/(3+3) = ½
r = TP/(TP+FN) =3/(3+2) = 3/5
F-measure = 2pr/(p+r)=6/11

TP+FN: the total number of positive samples
FP + TN: the total number of negative samples

Average Precision (1)

- Average Precision (AP):
  - –Averages precision over the entire range of recall
  - – A good score requires both high recall and high precision
  - – Application-independent



Average Precision (2)

Instance	P(+|A)	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.852	-
5	0.851	-
6	0.85	+
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

How to compute:

$$\frac{1}{5}\left(1+\frac{2}{2}+\frac{3}{6}+\frac{4}{8}+\frac{5}{10}\right)=0.7$$

How to Construct an ROC curve

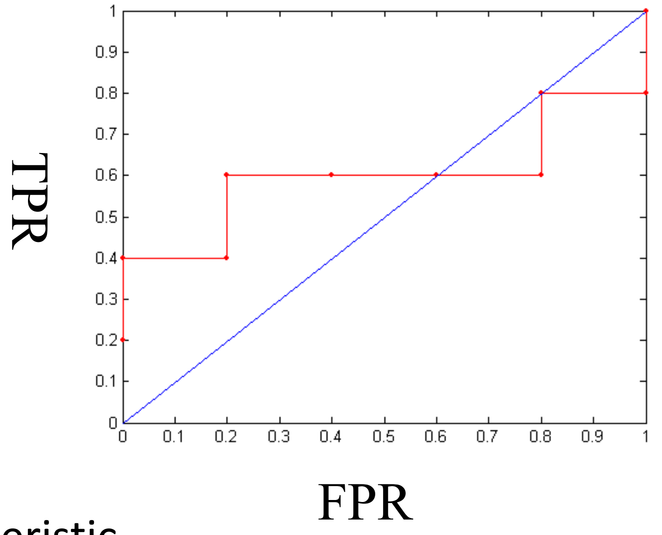
Instance	P(+|A)	True Class
1	0.95	+
2	0.93	+
3	0.87	-
4	0.852	+
5	0.851	-
6	0.85	-
7	0.76	-
8	0.53	+
9	0.43	-
10	0.25	+

- Use classifier that produces posterior probability for each test instance P(+|A)
- Sort the instances according to P(+|A) in decreasing order
- Apply threshold at each unique value of P(+|A)
- Count the number of TP, FP, TN, FN at each threshold
- TP rate, TPR = TP/(TP+FN)
- FP rate, FPR = FP/(FP + TN)

How to construct an ROC curve

Class	+	-	+	-	-	-	+	-	+	+	
Threshold >=	0.25	0.43	0.53	0.76	0.85	0.851	0.852	0.87	0.93	0.95	1.00
TP	5	4	4	3	3	3	3	2	2	1	0
FP	5	5	4	4	3	2	1	1	0	0	0
TN	0	0	1	1	2	3	4	4	5	5	5
FN	0	1	1	2	2	2	2	3	3	4	5
→ TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0
→ FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0

ROC Curve:



Receiver Operating Characteristic

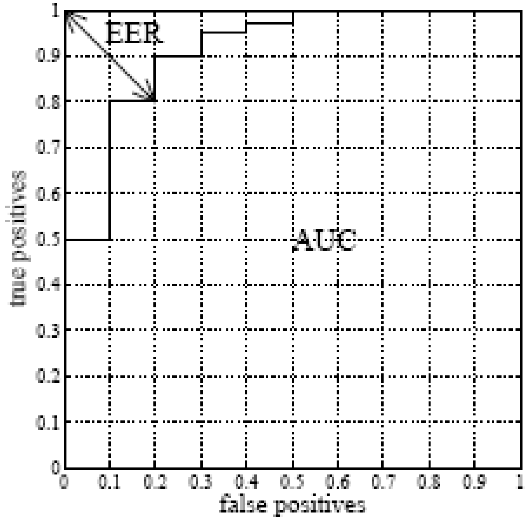
EER and AUC

- **Equal Error Rate (EER)** measures the **accuracy** at which the number of false positives and false negatives are equal.

  Somewhat emphasizes the behaviour of a method at low false positive rates which might be reasonable for a real-world application

- **Area Under Curve (AUC)** measures the total area under the ROC curve.

  Penalizes failures across the whole range of false positives, e.g. a method which recognizes some large proportion of instance with zero error but fails on the remaining portion of the data.



Thank you!